

Calculus A2 Homework

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1 HW Due Mar 9

Problem 1.1. You will get full credit for this problem as long as you submit this HW, no matter how you answer these questions.

1. What is your expected letter grade for this class?
2. How many hours per week do you plan to commit for this class?
3. Name a friend in this class you could discuss math with.
4. Name a friend that has already aced this class so you can ask questions from.
5. Name your favorite AI model as your personal assistant in all matters. (If you can afford it, use a paid version of whatever AI model. It is worth the investment in my opinion.)
6. No late HW will be accepted, unless you have doctor's notes or special permission from the university. Write "YES" for this subproblem to indicate that you understand.

Problem 1.2. You automatically get full credit for this problem.

1. Check out <https://tutorial.math.lamar.edu/ProblemsNS/CalcIII/CalcIII.aspx>. Alternative notes, exercises, and step-by-step solutions!
2. Check out <https://www.math.utoronto.ca/courses/mat237y1/20199/contents.html>. Alternative notes and exercises.
3. Check out <https://www.youtube.com/watch?v=TrcCbWwCBc&list=PLSQ10a2vh4HC5feHa6Rc5c0wbRTx56nF7>. Khan academy videos.
4. Check out <https://www.xiazr20.top/xiazr20/calculus-ii/.html#3>. This is a Chinese content, made by TA of past Tsinghua Chinese Calculus A2 classes by professor Xiaofeng Wang.
5. Check out [https://github.com/PKUanonym/REKCARC-TSC-UHT/tree/master//A\(2\)](https://github.com/PKUanonym/REKCARC-TSC-UHT/tree/master//A(2)). All Tsinghua exams in the past. However, warning. My exams will take a different style than most other professors.
6. Reflect on your own internet searching ability. Can you gather info/resources quickly and effectly about whatever subject, especially with the help of AI?
7. Finding the resources is only half the battle. USING the resources is the other half, and you should have a plan on how to use them. (But if you use multiple sources or info, beware of potential notation/definition differences and such. You should always standardize to our notation/definitions for our classs.)

2 HW Due Mar 16

Problem 2.1. Match the level curves and the graph of two-variable functions, and give a brief reason for each. (One sentence or two is fine.)

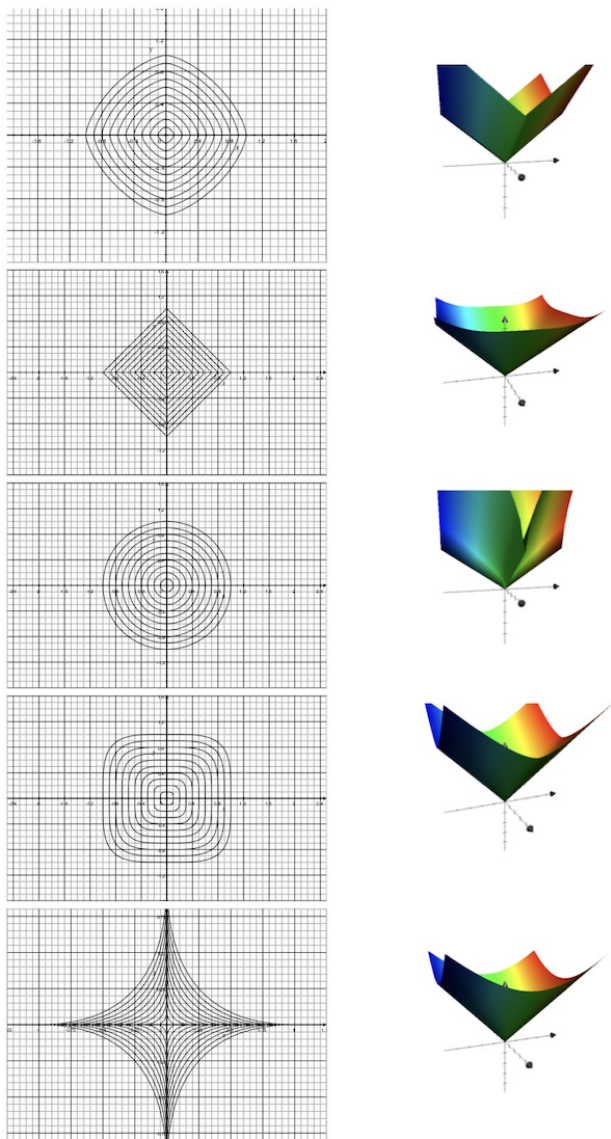
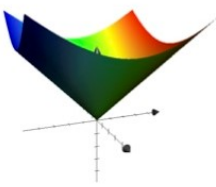
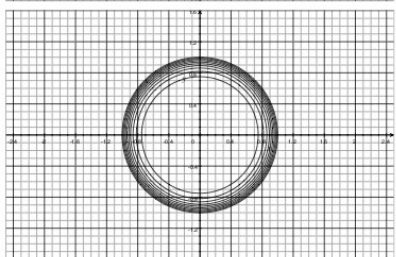
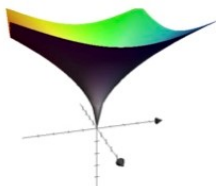
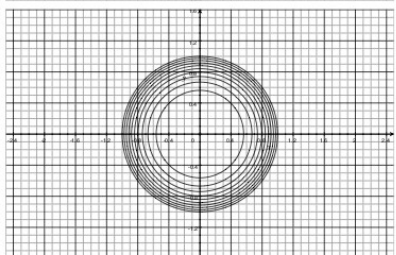
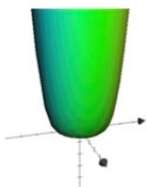
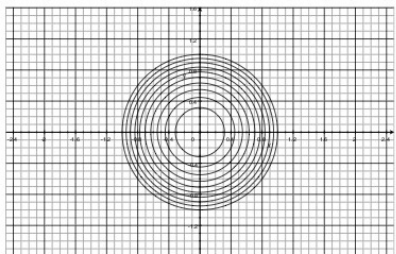
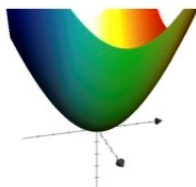
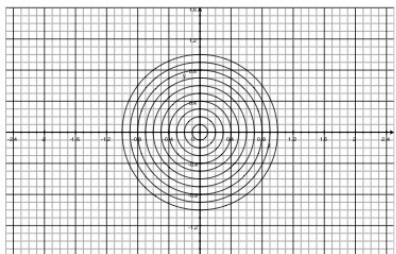
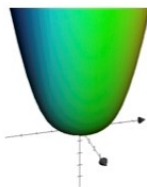
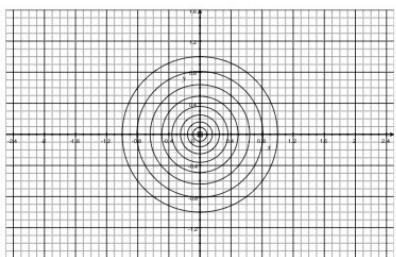
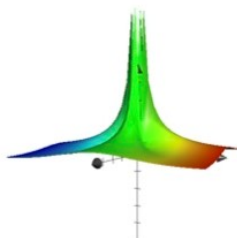
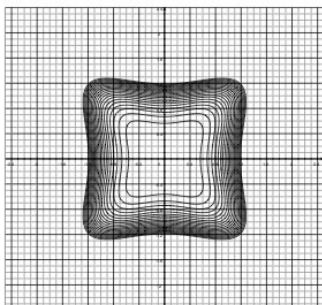
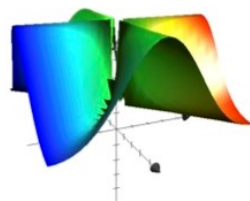
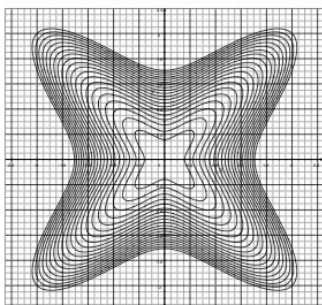
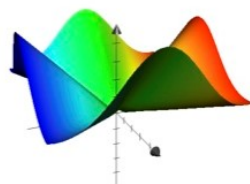
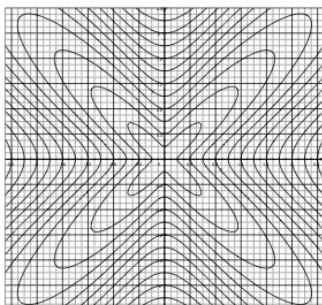
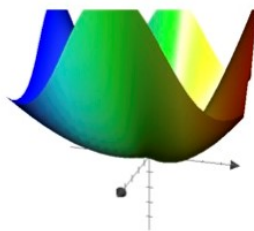
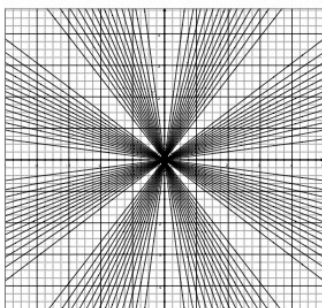
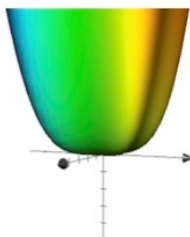
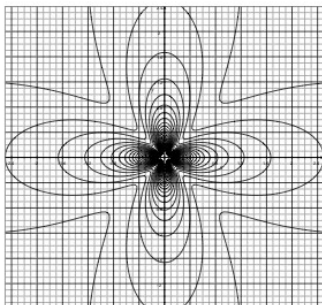


FIGURE 1. match the contour plots on the left with the graphs on the right

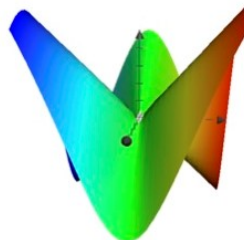
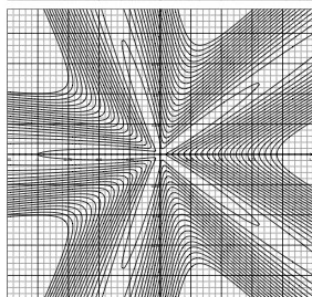
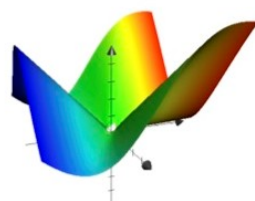
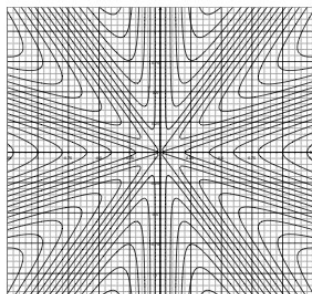
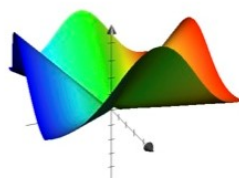
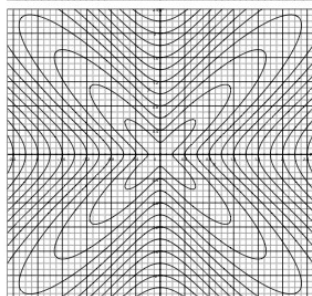
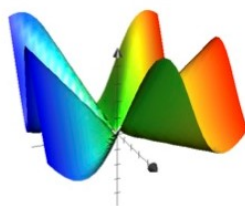
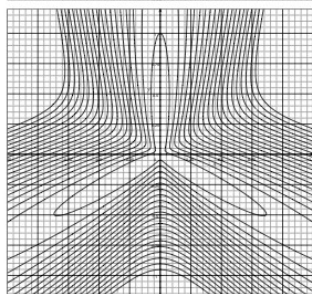
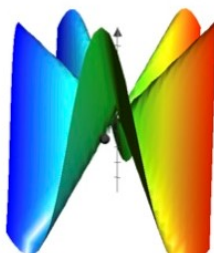
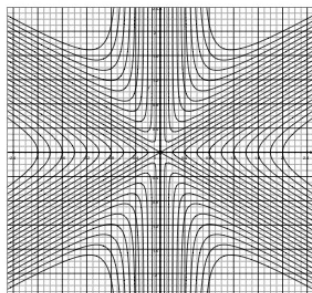
Problem 2.2. Match the level curves and the graph of two-variable functions. Density matters! And give a brief reason for each matching. (One sentence or two is fine.)



Problem 2.3. Match the level curves and the graph of two-variable functions, and give a brief reason for each. (One sentence or two is fine.)



Problem 2.4. Match the level curves and the graph of two-variable functions, and give a brief reason for each. (One sentence or two is fine.) Pay attention to the z -coordinates above and below the xy -plane.

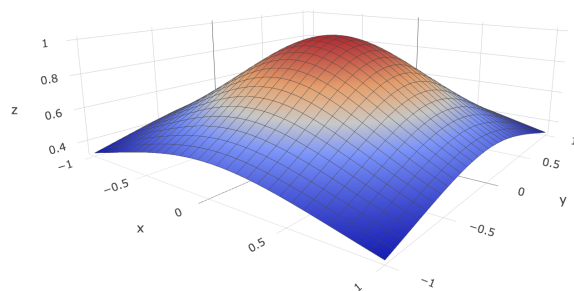


Problem 2.5. Which type of surfaces are these? (E.g., a sphere, an ellipsoid, a paraboloid, a plane, etc..) (The key is to look at the “slices” when you fix z , or maybe when you fix x , or maybe when you fix y .)

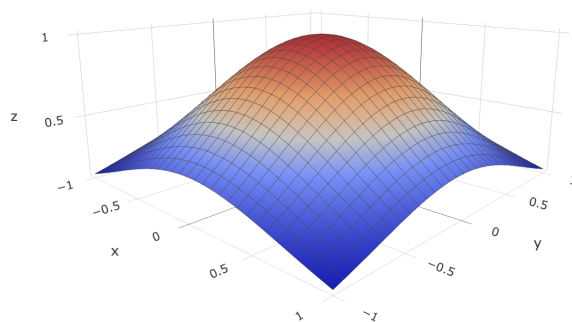
1. $x^2 - 4z - y = 2$.
2. $z^2 + 4x^2 = 1 - 4y^2$.
3. $z + 4x^2 = 1 - 4y^2$.
4. $z - 4x^2 = 1 - 4y^2$.

Problem 2.6. Match the functions and the graphs, and give a brief reason for each. (One sentence or two is fine.)

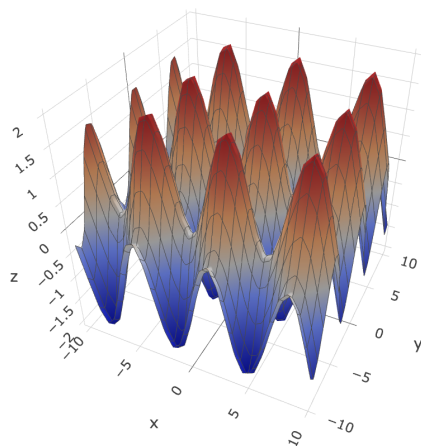
1. $f(x, y) = x^3 - 3xy^2$.
2. $f(x, y) = e^{-x^2 - y^2}$.
3. $f(x, y) = \sin x + \cos y$.
4. $f(x, y) = \frac{1}{x^2 + y^2 + 1}$.



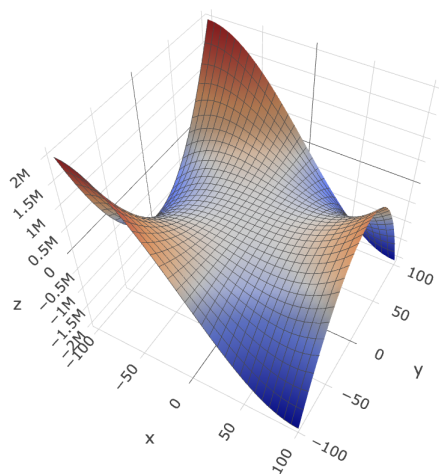
1. A bump.



2. A sharper bump.



3. Egg carton.



4. Monkey Saddle.

Problem 2.7. Write out the equations (in Cartesian coordinates) for the following surfaces in $\mathbb{R}^3$. The expression should only use POLYNOMIALS of x, y, z !

1. A sphere with center $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and radius 2.
2. An infinite cylinder around the line $x = y = z$ and radius 1.
3. A surface with cylindrical equation $z = r^2 \cos(2\theta)$. (Use the double angle formula.) This surface will go “up-down-up-down” as you travel around the z -axis.
4. A surface with cylindrical equation $z = r^3 \cos(3\theta)$. (Use the triple angle formula.) This surface will go “up-down-up-down-up-down” as you travel around the z -axis.
5. A surface with spherical equation $\rho \sin \phi = 2$.
6. A surface with spherical equation $\rho \cos \phi = 2$.
7. A surface with spherical equation $\rho \sin^2 \phi = \cos \phi$.

Problem 2.8. The following parametric expression describes the hypocycloid of a unit circle (b is a constant):

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (1-b)\cos t + b\cos(\frac{1-b}{b}t) \\ (1-b)\sin t - b\sin(\frac{1-b}{b}t) \end{bmatrix}.$$

Please plot this curve for $b = \frac{3}{10}$ with the help of whatever software. Is this a regular parametrization?

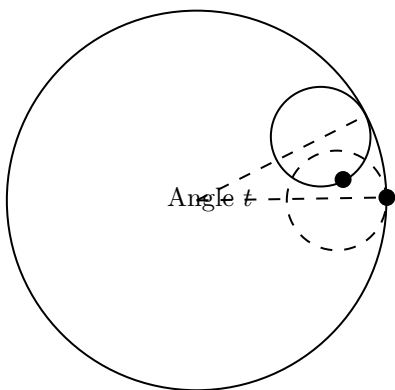
(Not part of the problem) For curious students, here is an explanation of why this represents a hypocycloid. Consider a unit circle, with a smaller inner circle of radius b rolling along its interior. The center of this inner circle rotates around a circle of radius $1-b$. Therefore, the trajectory of the center of the inner circle is:

$$\begin{bmatrix} (1-b)\cos t \\ (1-b)\sin t \end{bmatrix}.$$

On the other hand, assume that when the inner circle is at the far right (with its center at $\begin{bmatrix} 1-b \\ 0 \end{bmatrix}$), we mark the rightmost point of the inner circle. As the inner circle begins to roll and reaches the position where the center is $\begin{bmatrix} (1-b)\cos t \\ (1-b)\sin t \end{bmatrix}$, the arc length traveled along the contact interface between the two circles is exactly t .

This means the marked point on the inner circle has also traversed an arc length of t on the inner circle, which corresponds to an angle of $\frac{t}{b}$ within the inner circle. By subtracting this angular displacement from the inner circle's own rotation angle t , we obtain the angular position of the marked point as $-\frac{1-b}{b}t$. Adding the radius vector of the inner circle to its center position yields the final answer:

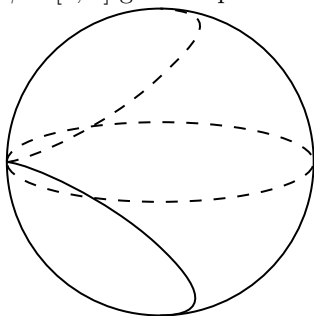
$$\begin{bmatrix} (1-b)\cos t + b\cos(\frac{1-b}{b}t) \\ (1-b)\sin t - b\sin(\frac{1-b}{b}t) \end{bmatrix}.$$



3 HW Due Mar 23

Problem 3.1 (Slowing Down on the Circle). Suppose there is a motion $\gamma : [0, \infty) \rightarrow \mathbb{R}^2$. Its initial position is $\gamma(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, and it rotates counterclockwise around the unit circle with a speed of $\|\gamma'(t)\| = e^{-t}$. Please write the expression for $\gamma(t)$, and then consider: does this motion ever complete one full revolution around the circle?

Problem 3.2 (Spiral on the Sphere). On the sphere of radius 1 (i.e., $\rho = 1$ always), the equation $2\phi = \theta$ for $\phi \in [0, \pi]$ gives a spiral curve on the sphere.



1. Find a parametrization of the curve using ϕ as a parameter.
2. Find a parametrization of the curve using θ as a parameter. Give me the parameter space explicitly!
3. Find a parametrization of the curve using z -coordinate as a parameter. (Note that $\cos$ and $\arccos$ and such in the expression could be further simplified!)
4. Calculate $\frac{d\theta}{dz}$.
5. Calculate the value of $\frac{ds}{d\theta}$ when $\theta = \frac{\pi}{3}$.
6. Calculate $(\frac{d\theta}{dz})(\frac{dz}{ds})(\frac{ds}{d\theta})$. (I tried asking Gemini, and Gemini gives me a brute force computation.... Surely you can do better than that?)

Problem 3.3 (Frenet Frame). Calculate the Frenet frame for the following curves, i.e., write $\mathbf{T}, \mathbf{N}, \mathbf{B}$ as functions of the parameter t . Also calculate the curvatures and torsion as a function of t . (Please simplify as much as possible.)

1. $\gamma(t) = \begin{bmatrix} \cos t \\ \sin t \\ t \end{bmatrix}$.
2. $\gamma(t) = \begin{bmatrix} 1 \\ t \\ t^2 \end{bmatrix}$.

Problem 3.4 (Frenet Decomposition of Acceleration). Please decompose the acceleration of the following curves according to the Frenet frame of the curve. That is, write $\gamma''(t)$ as a linear combination of $\mathbf{T}$ and $\mathbf{N}$. (Please simplify as much as possible.)

1. $\gamma(t) = \begin{bmatrix} 5 \cos(2t) \\ 5 \sin(2t) \\ 0 \end{bmatrix}$. This is a circle.

2. $\gamma(t) = \begin{bmatrix} 1+t \\ 1+2t \\ 1+3t \end{bmatrix}$. This is a straight line.

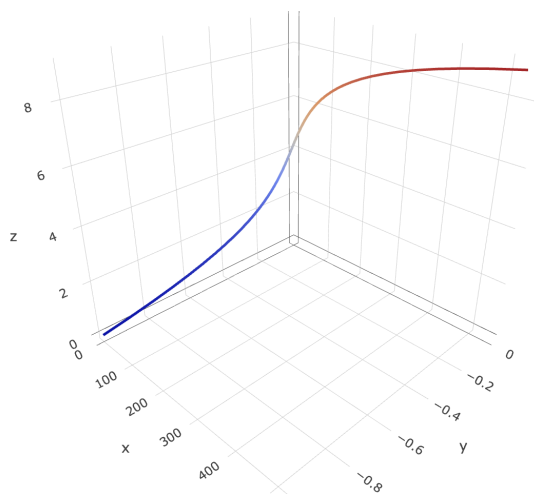
3. $\gamma(t) = \begin{bmatrix} t \cos t \\ t \sin t \\ 0 \end{bmatrix}$. This is a spiral.

4. $\gamma(t) = \begin{bmatrix} e^t \cos t \\ e^t \sin t \\ 0 \end{bmatrix}$. This is another spiral.

5. $\gamma(t) = \begin{bmatrix} 1 \\ t \\ t^2 \end{bmatrix}$.

Problem 3.5 (Last year midterm problem). I want to design a super awesome water slide. Note that this is the intersection curve of the two surfaces $xy + 1 = 0$ and $x = e^{\frac{z}{\sqrt{2}}}$. One possible parametrization is

$$\gamma(t) = \begin{bmatrix} e^t \\ -e^{-t} \\ \sqrt{2}t \end{bmatrix}.$$



1. Suppose we want to measure the length of the water slide. Compute the length of the curve for $t \in [-1, 1]$.
2. Compute the vectors $\mathbf{T}, \mathbf{N}, \mathbf{B}$ of the curve at the point $(1, -1, 0)$. (Hint: One method is to find $\mathbf{N}$ via the Frenet decomposition of the acceleration $\gamma''(t)$.)

Problem 3.6 (Integrations on curves). 1. Find the arclength parameter s as a function of t , for the curve

$$\gamma(t) = \begin{bmatrix} t^2 \\ 2t^3 \\ 1-t^3 \end{bmatrix}. \text{ We set } s = 0 \text{ when } t = 0.$$

2. Compute the indefinite integral $\int \mathbf{v}(t) dt$ where we have

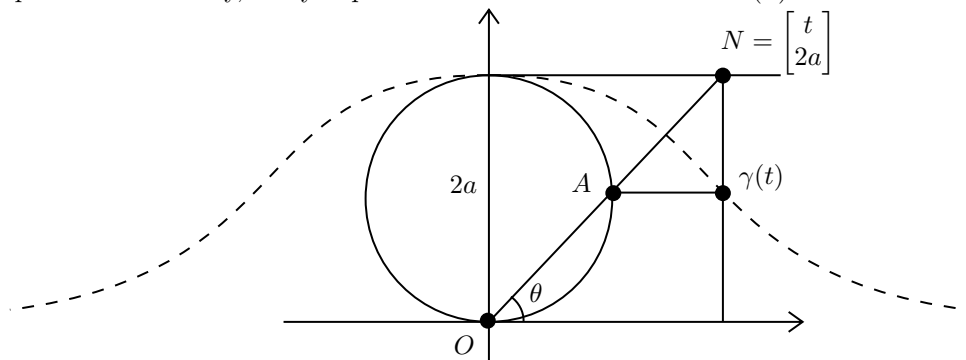
$$\mathbf{v}(t) = \begin{bmatrix} (1-t) \cos(t^2 - 2t) \\ \cos t \sin t \\ \sec^2(4t) \end{bmatrix}.$$

Keep in mind: should there be some “indetermined constants”? What would that looks like for this equation?

Problem 3.7 (♣Witch of Agnesi). Maria Agnesi was the first female mathematics professor in the world, and also the author of the first comprehensive calculus textbook (previously, authors wrote either only on differential calculus or only on integral calculus). In her calculus textbook, she discussed a class of curves now widely known as the Witch of Agnesi. (Agnesi originally named it the “versiera” or “rope that turns a sail,” but it was mistranslated into English, eventually becoming the “Witch” curve. However, the new name was so cool that it stuck, even inspiring a novel and a music album of the same name.)

Given a circle S with radius a and center $\begin{bmatrix} 0 \\ a \end{bmatrix}$. Note that the origin O is at the bottom of this circle. For any parameter t , we define the position of $\gamma(t)$ as follows: first, consider the point $N = \begin{bmatrix} t \\ 2a \end{bmatrix}$. Then, the line segment ON intersects the circle at point A . Let $y(t)$ be the y -coordinate of A , then define $\gamma(t) = \begin{bmatrix} t \\ y(t) \end{bmatrix}$.

Now, can you find the expression for $\gamma(t)$? Can you reparameterize this curve using the polar angle θ as the parameter? Finally, can you prove that the derivative of $\arctan(x)$ is a Witch of Agnesi curve?



4 HW Due Mar 30

Problem 4.1 (Open or Closed). Decide if the following subsets of $\mathbb{R}^n$ are open but not closed, closed but not open, both, or neither. If it is not closed, tell me which sequence has limit outside the subset. If it is not open, tell me which element can move infinitesimally out of the set. (Just point out the element and briefly describe the infinitesimal movement. No need to construct the infinitesimal explicitly.)

1. $S = \left\{ \begin{bmatrix} \frac{1}{n} \\ \frac{1}{n^2} \end{bmatrix} : n \in \mathbb{Z}^+ \right\}$. (Hint: Maybe plot a few elements to see better.)
2. $S = \{0\} \cup \left\{ \begin{bmatrix} \frac{1}{n} \\ \frac{1}{n^2} \end{bmatrix} : n \in \mathbb{Z}^+ \right\}$.
3. $S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : \left\| \begin{bmatrix} x \\ y \end{bmatrix} \right\| \leq 1, 2 \leq z \leq 3 \right\}$. (Hint: This is a Cartesian product of two subsets.)
4. $S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x < y < z \right\}$.
5. $S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x \leq y \leq z \right\}$.
6. $S = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x^2 \geq 0, y < 1 \right\}$.
7. $S = \{x \in \mathbb{R}^n : \|x\| \in (0, 1) \cap \mathbb{Q}\}$.
8. (No credit. Only if you want a challenge) The Cantor set C is constructed as such. We start with $C_1 = [0, 1]$. Removing its middle third portion, we get $C_2 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$. Then we remove the middle third portion of each interval, we get $C_3 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$. So on so forth. We define $C = \bigcap_{i=1}^{\infty} C_i$. (When I test this on Gemini, it sprouts some nonsense about how hard it is to show this is not open. Well... it is extremely easy to show this is not open.... I write this sentence to warn you to double check your AI answer.)
9. (No credit. Only if you want a challenge) $S = \{x \in \mathbb{R}^2 : \|x\| \in C\}$. Here C is the Cantor set above, and this set sort of looks like the ring of Saturn. (In reality, the ring of Saturn is using a “fat Cantor set” rather than the usual Cantor set, but the construction and idea are similar.)

Check out the website:

<https://galileo-unbound.blog/2023/11/28/of-fat-fractals-arnold-tongues-and-the-rings-of-saturn/>

You can ignore the words and the math, and simply scroll down to see the picture for fat Cantor fractal, and some pictures of the ring of Saturn at the end.

Problem 4.2 (Vector Sequence). Prove the following statements or give counter examples.

1. If $\lim a_n = a, \lim b_n = b$, then must we have $\lim(a_n \cdot b_n) = a \cdot b$? Prove or give counter example.
2. In $\mathbb{R}^3$, if $\lim a_n = a, \lim b_n = b$, must we have $\lim(a_n \times b_n) = a \times b$? Prove or give counter example.
3. I claim that the sequence $\left(\begin{bmatrix} \cos n \\ \sin n \end{bmatrix} \right)$ must have a converging subsequence. Prove this.

Problem 4.3 (Continuous Functions). Find the region on which the following functions are continuous. Give at least a brief argument for each. (But you should have an idea what a rigorous proof should look like.)

1. $f(x, y) = \frac{|x-1|^\pi |y-2|^{1+\pi}}{(x-1)^6 + (y-2)^8}$ with $f(1, 2) = 0$.
2. $f(x, y) = \frac{x^2 y^2}{(x^2 + y^2)^{\frac{3}{2}}}$ with $f(\mathbf{0}) = 1$.
3. $f(x, y) = \frac{x^2 + y^2}{x + y}$ with $f(x, y) = 0$ when $x + y = 0$.
4. $f(x, y) = (x^2 + y^2) \ln(x^2 + y^2)$ with $f(\mathbf{0}) = 0$.
5. $f(x, y) = (x^2 + y^2) \ln(x^2 + 2y^2)$ with $f(\mathbf{0}) = 0$.
6. $f(x, y) = \frac{\sin(x+y)}{\sqrt{x^2 + y^2}}$ with $f(\mathbf{0}) = 1$.
7. $f(x, y) = \frac{\sin(x^3 + y^3)}{x^2 + y^2}$ with $f(\mathbf{0}) = 0$.
8. $f(x, y) = \frac{x}{y^2} e^{-\frac{x^2}{y^2}}$ when $y \neq 0$, and $f(x, y) = 0$ when $y = 0$.
9. A triangle on the plane is determined by its three vertices, each with a pair of coordinates. Hence the “triangle space” is $\mathbb{R}^6$, where an element $\begin{bmatrix} \mathbf{a} \\ \mathbf{b} \\ \mathbf{c} \end{bmatrix}$ for $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^2$ represents the triangle with vertices $\mathbf{a}, \mathbf{b}, \mathbf{c}$. Our map $f : \mathbb{R}^6 \rightarrow \mathbb{R}$ is sending each triangle to its area. (Don’t find any formula. Argue with geometric intuition.)
10. f sends each invertible matrix to its inverse. (You may assume the matrix is 2×2 for simplicity.)
11. f sends each point on $\mathbb{R}^2 - \{\mathbf{0}\}$ to θ in its polar coordinate, where $\theta \in [0, 2\pi)$.

Problem 4.4 (Hints to the last problem). Obviously you don’t need to do anything. Feel free to look at these hints, or don’t look at them if you want a challenge.

1. Substitute $u = x - 1, v = y - 2$, then this turns into the class example.
2. Polar Coordinates.
3. At the origin, try a parabola tangent to the line $x + y = 0$.
4. Polar Coordinates.
5. Easier than you think. Surely you know $x^2 \leq 2x^2$.
6. Try some directional limits.
7. Divide and multiply by $x^3 + y^3$.
8. At the origin, try $x = y$ versus $x = y^2$.
9. Obvious.
10. Obvious from the big formula for inverse matrix.
11. What happens at $(1, 0)$ when we move downward infinitesimally?